Gourab Saha

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Research Summary

Electrical and electronic engineering graduate working on renewable energy optimization, biomedical signal processing, and IoT-based sensing. First author of three Q1 journal articles, co-author of six conference papers, and journal reviewer, with two manuscripts under review and undergraduate teaching experience. Seeking MS/PhD research in smart energy systems, IoT sensing, and precision agriculture.

Research Interests

Smart Grids and Power Systems, Control Systems, Cyber-Physical Security, Applied AI and Edge Computing, Remote Sensing and IoT, Precision Agriculture

Education

B.Sc. in Electrical and Electronic Engineering May 2019 – Oct 2023
BRAC University, Dhaka, Bangladesh
CGPA: 3.74/4.00 (Graduated with High Distinction)

Relevant Coursework: Control Systems, Power Electronics, Energy Conversion, Digital Signal Processing, Microprocessors, VLSI Design, Semiconductor Devices and Materials, Numerical Methods, Electromagnetic Waves and Fields

Research Experience

Research Assistant, BRAC University Sep 2024 – Present
Supervisor: Prof. A.K.M. Abdul Malek Azad
Brain-Computer Interfaced Wheelchair for Quadriplegia Patients

- Developed a real-time blink-intent detection pipeline using EMG, with a cascaded Butterworth band-pass and 50 Hz notch filter for artifact removal. Compared EMG with single-channel EEG and found EMG more reliable for blink-based control, reaching 93% accuracy at 0.45 s average latency across 15 participants.
- Designed a scanning-based control interface, implemented in real-time firmware on Arduino and ESP platforms, that lets users with severe motor impairment drive the wheelchair with a single intentional blink.
- Built an obstacle avoidance layer and compared ultrasonic and time-of-flight (ToF) sensors, with ToF showing 13% lower response time.

Research Assistant, BRAC University Nov 2024 – May 2025
Supervisor: Dr. A.S. Nazmul Huda
Demand Side Management and Optimal Sizing of a Hybrid Microgrid for an Industrial Load

- Collected machine-level load data from a garment-accessories factory (1.44 MW peak) and classified 40 loads as critical, shiftable, or clippable to build a realistic industrial load profile.
- Modeled load shifting, peak clipping, and a combined strategy for a PV–Wind–Diesel–Battery microgrid in HOMER Pro, with stochastic load variability (10% day-to-day, 20% time-step) to reflect production uncertainty.
- The combined strategy cut the daily electricity bill by over 30% and Net Present Cost (NPC) by about 18%, raised the renewable fraction from 57.8% to 71.9%, and reduced fuel use and CO_2 emissions by about 30%, with robustness confirmed through multi-parameter sensitivity analysis.
- Formulated sizing as a constrained NPC minimization (LPSP ≈ 0) and implemented PSO, GA, and DE, which converged within 1% of each other. GA found a design 7.58% cheaper than HOMER ($32.67M NPC), with about 3,600 kWh less battery storage and a 44% smaller diesel generator.

Research Assistant, BRAC University Oct 2023 – Sep 2024
Supervisor: Prof. A.K.M. Abdul Malek Azad
IoT-Driven Precision Agriculture: Land Mapping, Crop Prediction, and Irrigation Control

- Mapped vegetation and soil salinity using NDVI/NDSI and K-means on Landsat-8 imagery (2015–2019), and forecast seasonal vegetation area with an LSTM model.
- Built an ESP32-based soil sensing unit linked to a cloud ML pipeline. Selected Random Forest from eight classifiers (97.35% accuracy across 22 crops) and validated it on soil from three districts and in an outdoor field test.
- Designed a solar-powered fuzzy-logic irrigation controller that used about 70% less water and ran 66% faster than a threshold-based controller in bench tests.

Publications

Journal Articles (Published)

- **G. Saha**, M. T. Imran, S. M. Shams, M. F. A. Khan, A. Malek, and M. A. Imteaz, "From air to water: Science, technology, and future of atmospheric water harvesting (AWH)," *Applied Thermal Engineering*, vol. 302, part 5, 2026.
- **G. Saha**, F. Shahrin, F. H. Khan, M. M. Meshkat, and A. Malek, "Smart IoT-driven precision agriculture: Land mapping, crop prediction, and irrigation system," *PLoS ONE*, vol. 20, e0319268, Mar. 2025.
- **G. Saha** and A. Malek, "A review of advancements in solar PV-powered refrigeration: Enhancing efficiency, sustainability, and operational optimization," *Energy Reports*, vol. 12, pp. 1693–1709, Dec. 2024.

Manuscripts Under Review

- **G. Saha**, F. A. Nafis, M. H. A. Araf, S. M. Shams, and A. K. M. A. Azad, "A comprehensive review of brain-controlled wheelchairs: Recent advancements in EEG, EOG, hybrid technologies, and artifact removal techniques for mobility control," *Heliyon*, 2025 (revision under review).
- A. R. Sagor, **G. Saha**, S. H. Suvo, S. Ahmad, and A. S. N. Huda, "Optimal sizing of hybrid power system using metaheuristic optimization algorithms considering demand side management," *Energies*, 2025 (under second-round review).

Conference Papers

- F. A. Nafis, S. S. Alvi, S. T. S. Rafid, N. Akib, A. J. Anonna, and **G. Saha**, "A reliable robotic solution for hospital logistics: Implementation of intelligent pathfinding and control strategy," *ECCE*, Chittagong, Bangladesh, 2025, pp. 1–6.
- S. S. Alvi, F. A. Nafis, N. Akib, S. T. S. Rafid, A. J. Anonna, and **G. Saha**, "Voting and stacking-based ensemble methods to detect Bengali spam SMS with proposal for comprehensive dataset," *ECCE*, Chittagong, Bangladesh, 2025, pp. 1–6.
- F. A. Nafis, N. Akib, **G. Saha**, S. S. Alvi, M. I. A. Anik, and A. J. Anonna, "An efficient IoT-based fire detection system using quantized deep learning model on resource-constrained devices," *ICCIT*, Cox's Bazar, Bangladesh, 2024, pp. 3182–3187.
- **G. Saha**, S. H. Suvo, F. T. Tonmoy, J. B. Asad, M. T. Imran, and A. A. M. Azad, "Smart soil monitoring system with crop and fertilizer recommendation features," *CSNT*, Jabalpur, India, 2024, pp. 1190–1196.
- M. A. H. Faiyaz, **G. Saha**, S. H. Suvo, and S. T. S. Rafid, "Renewable energy-driven microgrid design for a desalination plant in Noakhali, Bangladesh," *CSNT*, Jabalpur, India, 2024, pp. 1330–1335.
- S. H. Suvo, Nur-E-Tasniya, T. U. Wara, **G. Saha**, Z. Alam, and M. Islam, "Energy prospects of floating solar PV considering cloud cover: An analysis on Hatirjheel Lake," *ICEEICT*, Dhaka, Bangladesh, 2024, pp. 1146–1151.

Teaching Experience

Adjunct Lecturer, Dept. of EEE, BRAC University — Jun 2025 – Jan 2026

- Taught and assessed four lab sections of *Electrical Circuits I* and *Digital Signal Processing*, with over 130 students.

Teaching Assistant, Dept. of MNS, BRAC University — Sep 2022 – Sep 2023

- Delivered tutorials and mentored undergraduates in *Elements of Statistics and Probability*, and managed course grading and assessments.

Student Mentor, Office of Academic Advising, BRAC University — May 2022 – Sep 2023

- Mentored first-year EEE students and ran tutorial sessions to help them improve academically.

Teaching Interests: Electrical Circuits, Digital Signal Processing, Control Systems, Power Electronics, Embedded Systems

Technical Skills

Programming: Python, MATLAB, Embedded C, C++
Power and Energy Tools: HOMER Pro, PVsyst, MATLAB/Simulink, OpenDSS, GridLAB-D
Optimization and Control: PSO, GA, DE, fuzzy logic control
Signal Processing: digital filter design, EMG/EEG processing
Embedded Systems and IoT: ESP32, Arduino, Raspberry Pi, MQTT, Modbus
Machine Learning and Data: Scikit-learn, PyTorch, OpenCV, Pandas, NumPy, GeoPandas
Circuit and Hardware Design: PSpice, Proteus, Cadence, KiCad, SolidWorks
Remote Sensing and GIS: ArcGIS, Google Earth Engine

Professional Service

Reviewer, *Journal of Automation, Mobile Robotics and Intelligent Systems* — 2025 – Present
Reviewer, *International Transactions on Electrical Energy Systems* — 2026 – Present

Honors and Awards

- Rewarding Quality and Publication (RQJP) Award, BRAC University — 2026
- Graduated with High Distinction, BRAC University — 2023
- Vice Chancellor's List (two terms) and Dean's List (one term), BRAC University — 2019 – 2023